MUHAMMED SALIM K T

Senior AI/ML Engineer | Python (LLM, RAG, AI Agents & Speech AI)

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Experience: 4+ years (production AI & Python) **Work mode:** Remote, hybrid or on-site
Open to: Remote (global) or relocation; visa sponsorship needed outside India **Notice period:** 30 days
Time zones: Comfortable overlapping with US, EU and Gulf hours

PROFESSIONAL SUMMARY

AI/ML Engineer and senior Python developer with 4+ years of building and running production systems on **FastAPI, Django, Celery, Redis and PostgreSQL**. Built a production **speech-AI service** for a US telephony provider (NVIDIA Canary and Whisper on vLLM), a **multi-agent tool-calling engine** that lets a 12B model do 32B-class work, and **RAG over 10M+ embeddings** serving 10,000+ queries a day. Experienced in on-premise and air-gapped **Kubernetes** delivery for security-sensitive clients, LLM evaluation gates in CI, and working directly with US enterprise clients. Open-source contributor to **Celery** and **PaddleOCR**.

10M+ embeddings | **10k+** queries/day | **99.7%** uptime | **12B → 32B**-class work | **50k+** concurrent users

CORE SKILLS

Languages & Backend: Python, FastAPI, Django, Flask, Celery, asyncio, Pydantic, REST APIs, WebSockets, SSE, Node.js, TypeScript, RabbitMQ, Kafka, Microservices

Generative AI / LLM: LLM applications, RAG, hybrid search and re-ranking, AI agents, tool calling, MCP, LangChain, LlamaIndex, prompt engineering, LLM evaluation and guardrails, human-in-the-loop workflows, n8n

Machine Learning: PyTorch, Hugging Face Transformers, fine-tuning (LoRA / QLoRA, PEFT, Unsloth), NLP, scikit-learn, pandas, NumPy, anomaly detection

Speech & Vision: Speech-to-text (Whisper, NVIDIA Canary), speaker diarization (SpeechBrain, Silero VAD), text-to-speech, OCR (PaddleOCR, RapidOCR, Docling), LayoutLMv3, Qwen-VL

Model Serving: vLLM, Ollama, OpenAI-compatible APIs, quantization, batching, caching, failover, on-premise / air-gapped deployment

Data & Vector DBs: PostgreSQL, pgvector, Qdrant, Cassandra, FAISS, Redis, MongoDB, MySQL, DuckDB, SQLAlchemy, ETL

Cloud & DevOps: Kubernetes, Docker, GCP (Cloud Run, Vertex AI), GitLab CI, GitHub Actions, Prometheus, Grafana, Linux, Git, Agile / Scrum

Frontend & Mobile: React, React Native, TypeScript (Web, Android & iOS)

PROFESSIONAL EXPERIENCE

AI/ML Engineer & Full-Stack Developer (Python) — Jun 2022 – Present

Netstratum Technologies Pvt Ltd, Kochi, India · Python lead for an on-premise enterprise AI platform and speech / real-time products for US enterprise clients.

SPEECH AI: ECHO (PRODUCTION)

- Built **Echo**, a production speech-AI service that transcribes voicemails and call recordings for a **US hosted-telephony provider**, using **NVIDIA Canary and Whisper on vLLM** with automatic failover between engines.
- Designed a fault-tolerant pipeline: **Redis priority queues** with **Celery** workers on stateless HA nodes, output validation (not just status codes), duplicate suppression and a GPU-error watchdog.
- Built long-audio processing (recordings up to 10 hours), language detection, **speaker diarization** with SpeechBrain, and LLM summaries and keywords on a separate queue so they never block transcription.

LLM, RAG & AGENTIC AI

- Built an **on-premise enterprise AI chat platform** in Python / FastAPI serving **10,000+ queries a day**, with no customer data leaving the client's infrastructure.
- Designed RAG end to end: Docling + RapidOCR ingestion, embeddings in **Qdrant**, metadata in Cassandra, and hybrid keyword + semantic search with re-ranking over **10M+ embeddings**. Improved answer accuracy **45%** over the base LLM and cut retrieval time **70%**.
- Cut average inference latency to **1.8 s** with quantization, batching and caching; held **99.7% uptime** with model failover.

- Authored a **multi-agent tool-calling engine** that replaced CrewAI in production. Strict tool schemas and orchestration in code let a **12B model do 32B-class work**, moving customer deployments from a GPU cluster to a **single DGX node**.
- Built a tool registry of **73 toolsets** (SharePoint / MS Graph, GitLab, Jira, BigQuery, Snowflake, Kafka) and a model router (small / large / reasoning tiers) over vLLM and Ollama, with streaming responses.
- Fine-tuned Hugging Face models, including **Arabic LLMs with LoRA / QLoRA**, and built an evaluation framework that gates every release with a regression check.

REAL-TIME, DOCUMENT AI & PLATFORM

- Built Python and Node.js WebSocket / REST services for **50,000+ concurrent users**, adding speech-to-text, text-to-speech and live captions to a video-calling product.
- Built a document-AI service (PaddleOCR-VL, Qwen-VL) and fine-tuned **LayoutLMv3 to 91% F1** for invoice and contract extraction on 100K+ documents a month.
- Own **Kubernetes** delivery across **8 environments, including two GPU/DGX clusters**, plus GCP Cloud Run and GitLab CI pipelines. Took the platform through client security reviews (encryption, RBAC, audit logging).
- Built ML log analysis (Isolation Forest + LSTM) over **5M+ events a day**, cutting incident detection time **60%**.
- Delivered for US enterprise clients, including a video-conferencing product for **Windstream Enterprise** and a mobile app with **50,000+ downloads**.

SELECTED PROJECTS & OPEN SOURCE

DataRat: AI Data Analyst (Python, FastAPI, DuckDB, pandas, React). Four AI agents (Planner, SQL, Analyst, Narrator) answer plain-English questions over 11 kinds of data source, with anomaly detection, forecasting and a read-only SQL guard. 122 automated tests.

small-model-agents (Python, MCP, GitHub Actions). Reference implementation of reliable agents on small models: schema-validated tool calling, model routing and a CI evaluation gate.

AI ticket triage with human approval (n8n, LLM, structured outputs). An LLM classifies support tickets and drafts replies; coded guardrails send refunds, warranty, low-confidence and high-value cases to a human Approve / Edit / Reject step. Tested with Arabic and prompt-injection cases.

3xDezine: home design and costing platform (TypeScript, Fastify, PostgreSQL, WebGPU). 2D floor plan to 3D walkthrough with live material estimates and quotations; 300+ unit tests and payment billing with idempotent webhooks.

Open-source contributions: Celery PR #10968, logging for recycled prefork workers with tests and docs; **PaddleOCR** PRs #18416–18418, fixes to its MCP server and Word-to-Markdown conversion; design proposals to **vLLM** and **Silero VAD**.

EDUCATION

B.Tech, Computer Science & Engineering
KMEA Engineering College (APJ Abdul Kalam Technological University), Kerala, India
2018 – 2022 · CGPA 7.53 / 10

CERTIFICATIONS

- Fundamentals of Deep Learning: NVIDIA Deep Learning Institute
- Architecting with Google Compute Engine: Google Cloud (Coursera)
- Robotic Process Automation: UiPath

LANGUAGES

- English: Fluent
- Arabic: Intermediate
- Malayalam: Native
- Hindi, Urdu